Safer, Not Slower

Actionable insights from a large-scale survival analysis of baseball pitching mechanics

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Conference on Complex Systems 2026
Complexity & Sports Satellite



RICE UNIVERSITY
Sport Analytics

Collaborators

Rose Graves
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Reboot Motion

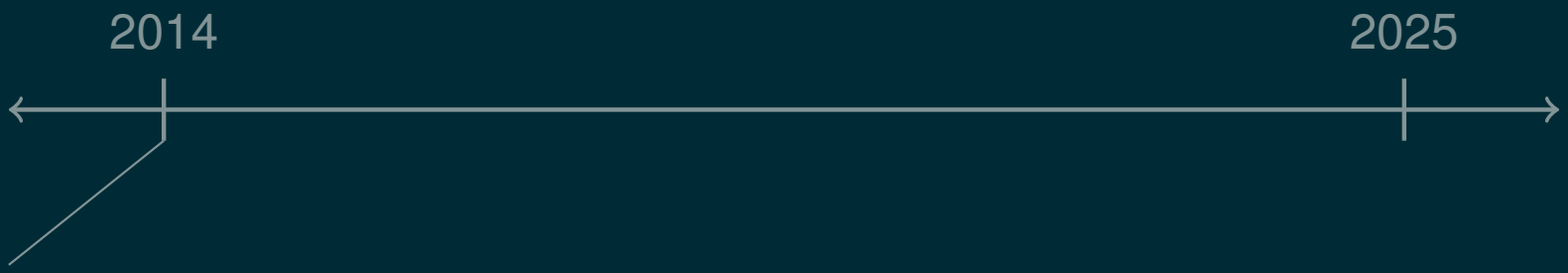
Daniel Barrueco
MLB

John D'Angelo
MLB

Amanda Glazer
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. . . and many others!

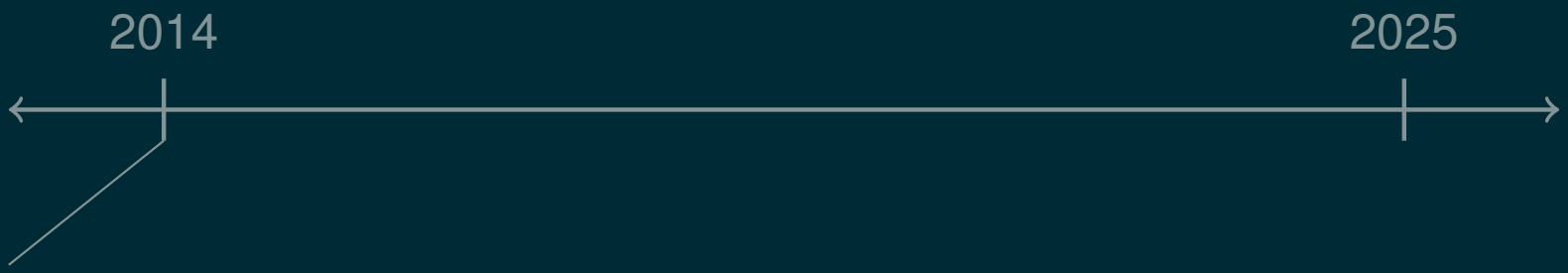
The Problem



"From my standpoint, [Tommy John surgery] is an **epidemic**."

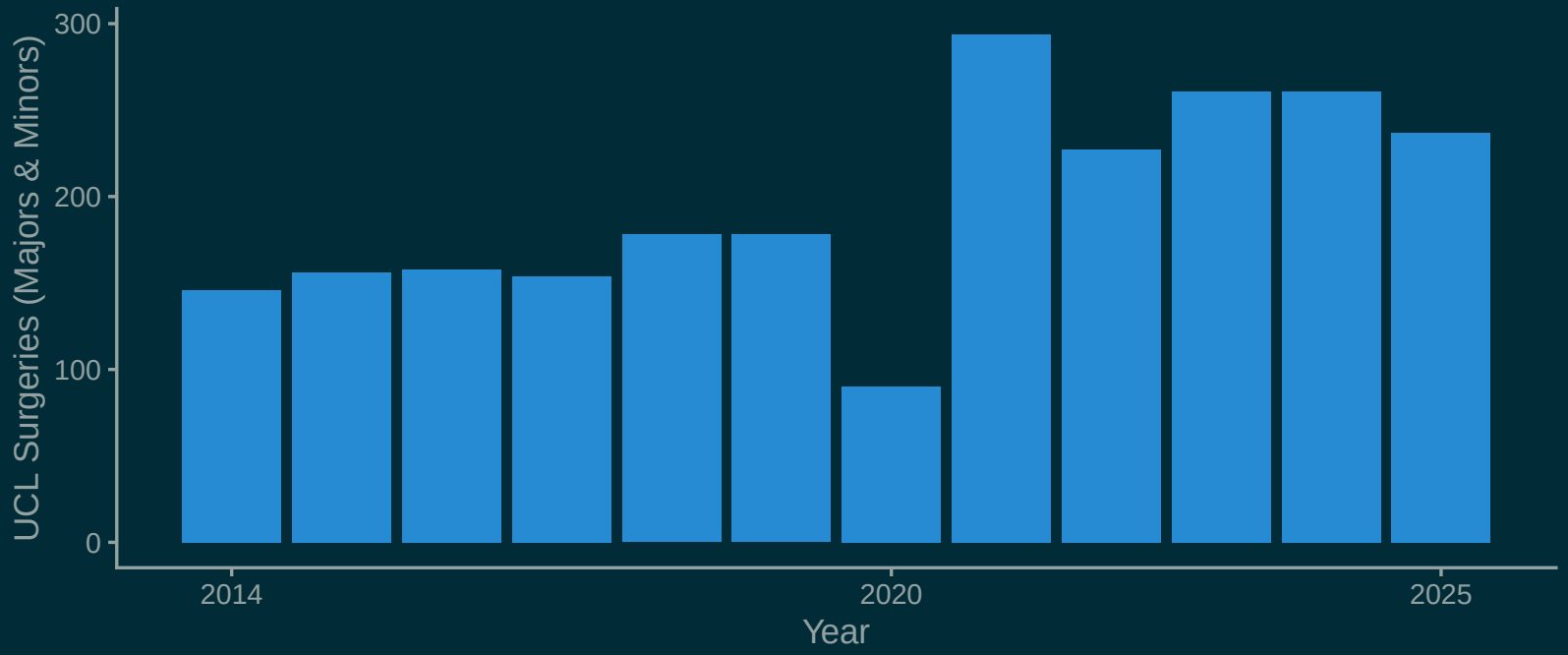
— Dr. James Andrews

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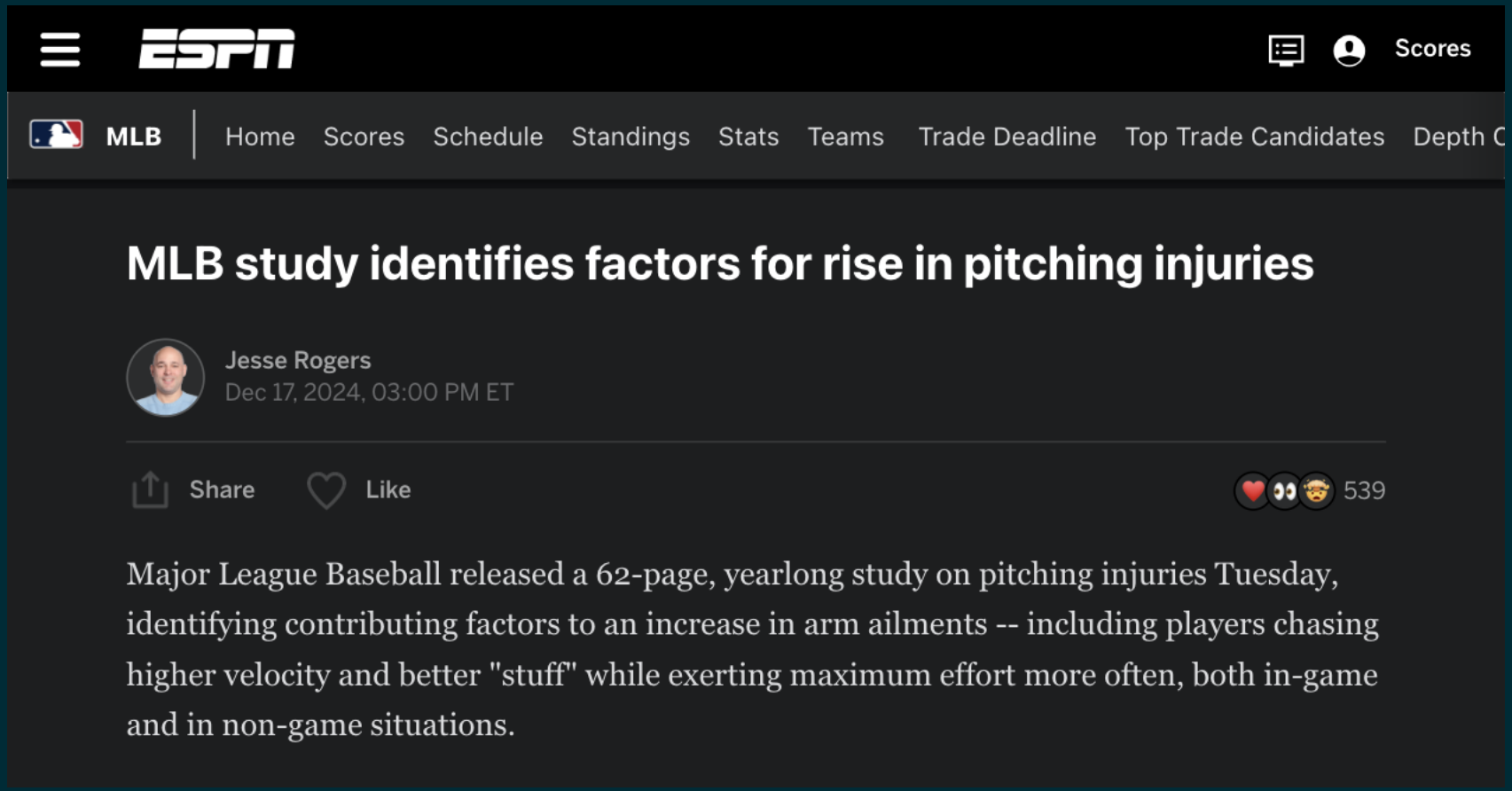


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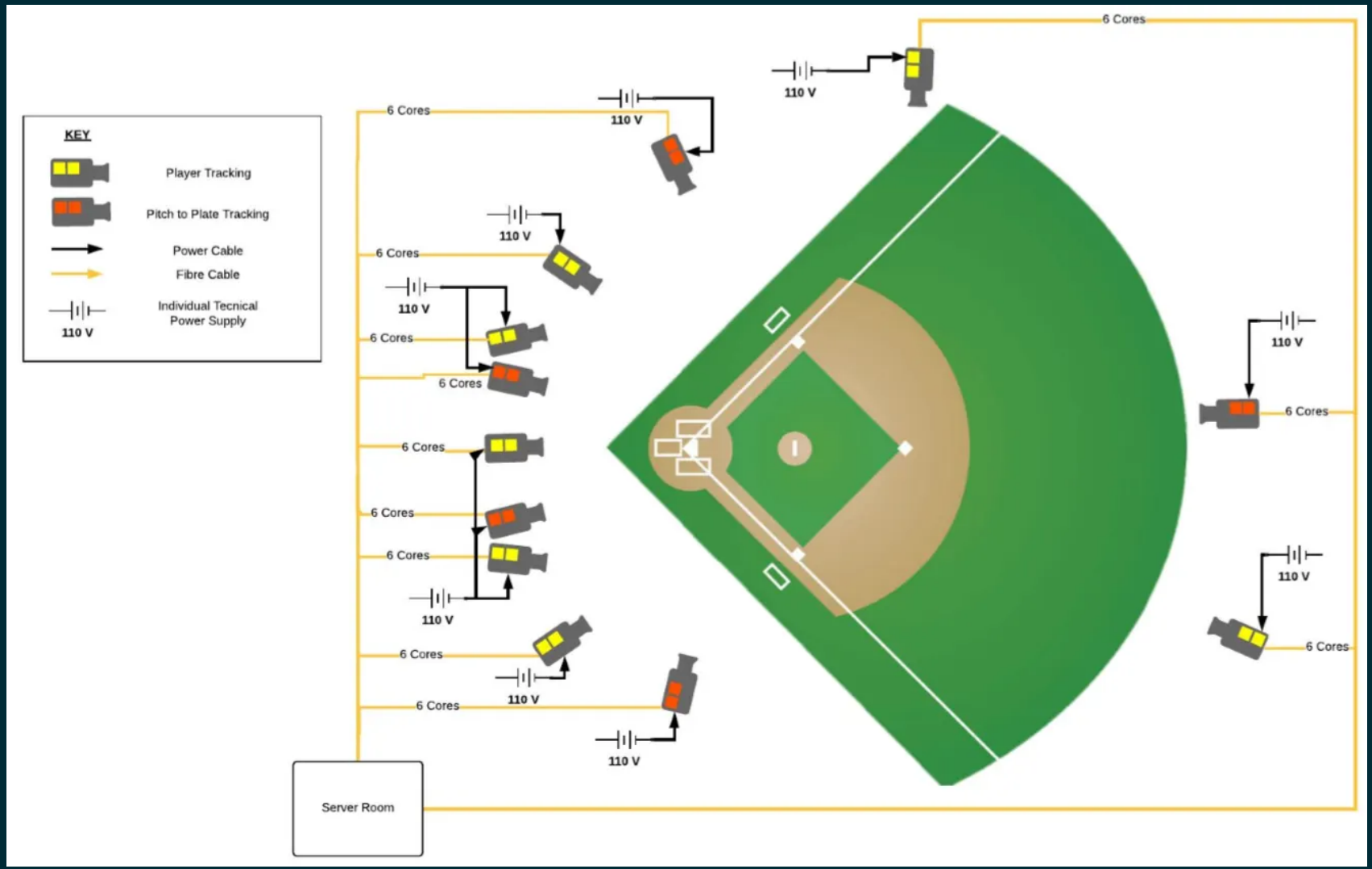
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2024 MLB Report



Data Generation



Data

Pitch-by-pitch:

- ball tracking, MLB 2017+ (5,972,263 pitches)
- ball tracking, minors 2021+ (11,322,383 pitches)
- **skeletal pose** tracking, MLB 2023+ (1,130,824 pitches)

Injury outcomes

- 8,234 throwing-arm IL placements, MLB & minors 2018+
- 1,726 (reported) UCL surgeries, MLB & minors 2018+

IL = injured list

UCL = ulnar collateral ligament

Methods

Framework:
year-Y features $\longmapsto$ year-$(Y+1)$ pitches until IL

Model: multivariate proportional hazards regression*

Cohorts:

- Majors 2018–2025 (3,828 pitcher-seasons)
- Minors 2022–2025 (6,241 pitcher-seasons)
- **Biomechanics 2024–2025** (1,211 pitcher-seasons)

*clustered standard errors account for repeated pitcher observations

Pitch-Tracking Features

Demographics

- Hand (L/R)
- Role (SP/RP)
- Past Injury (0/1)
- Age

Pitch Types

- SI Usage
- CT Usage
- SL+SW Usage
- CB+KC Usage
- CH+SF Usage

Pitch Tracking

- Primary Velo
- Primary Spin Rate
- Lateral Release Location
- FB Vert Break*
- SL Spin Rate*
- CB Spin Rate*
- Glove-Side Break
- Arm-Side Break
- Stuff+
- **Primary Velo Effort Ratio**

*interacted with usage

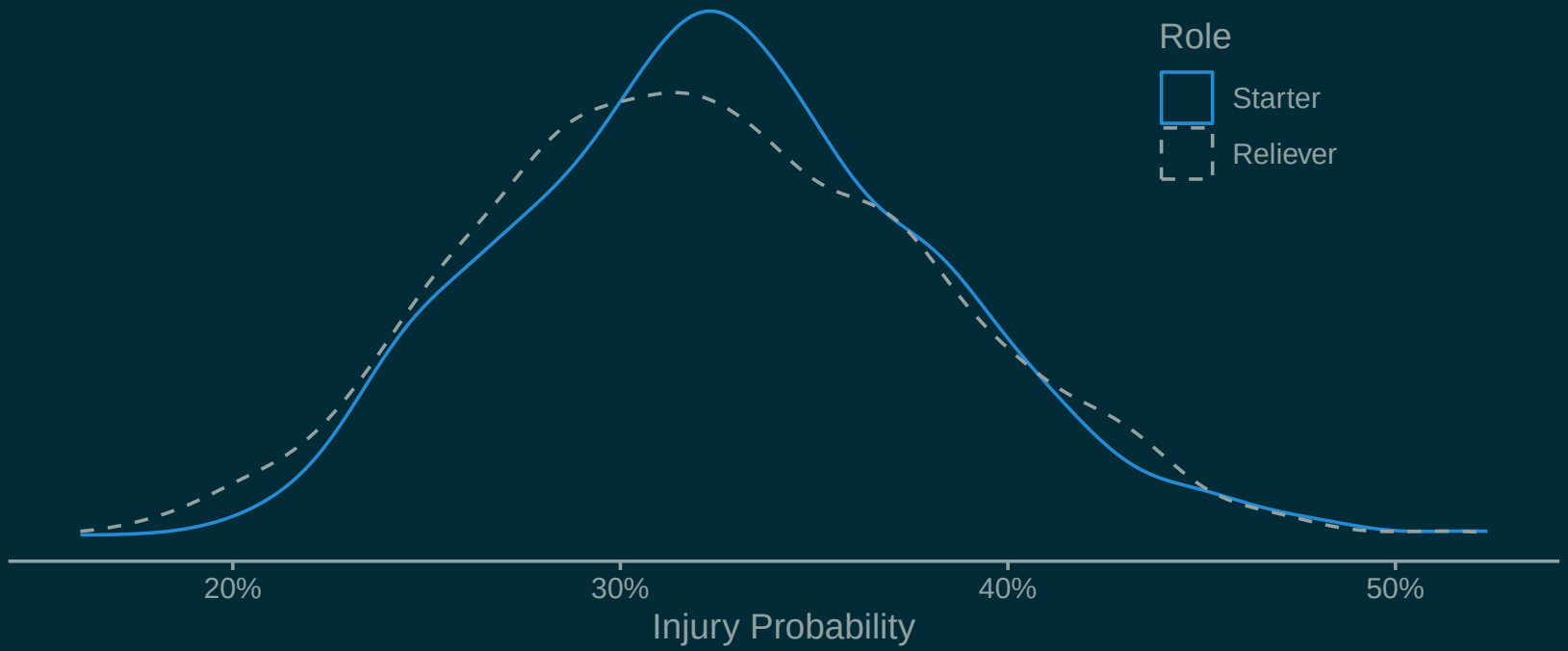
Pitch-Tracking Results

	95% Conf. Int.
Majors 2018–2025 cohort	
• 1 mph Primary Velo $\leftrightarrow +5\%$ hazard	$(+2\%, +8\%)$
• **10pp SL+SW Usage $\leftrightarrow +8\%$ hazard**	$(+1\%, +15\%)$
• 1 SD FB Vert Break $\leftrightarrow +10\%$ hazard	$(+3\%, +18\%)$
Minors 2022–2025 cohort	
• 1 mph Primary Velo $\leftrightarrow +5\%$ hazard	$(+2\%, +8\%)$
• **10pp SL+SW Usage $\leftrightarrow +6\%$ hazard**	$(+0\%, +13\%)$
• 10pp CH+SF Usage $\leftrightarrow +8\%$ hazard	$(+1\%, +16\%)$
• 1 SD Lat. Rel. Position $\leftrightarrow -7\%$ hazard	$(-12\%, -2\%)$

SL+SW = slider/sweeper
CH+SF = changeup/splitter

pp = percentage point
SD = standard deviation

Compounding Risk



Fitted injury probability* **varies greatly** across pitcher-seasons

*Reflects 2,500-pitch starter workload or 1,000-pitch reliever workload

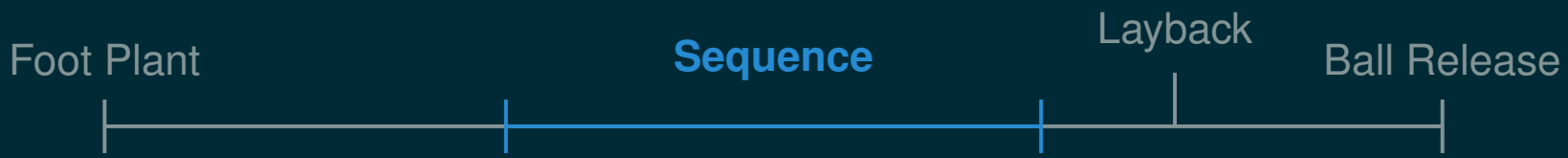
Biomechanics Features (Joint Angles)

Joints:

- Rear Hip (3 df)
- Lead Knee (1 df)
- Lumbar (3 df)
- Shoulder (3 df)
- Elbow (1 df)

Summary stats:

- min/max
- min/max timing
- min/max in **sequence**
- range utilization
- at layback

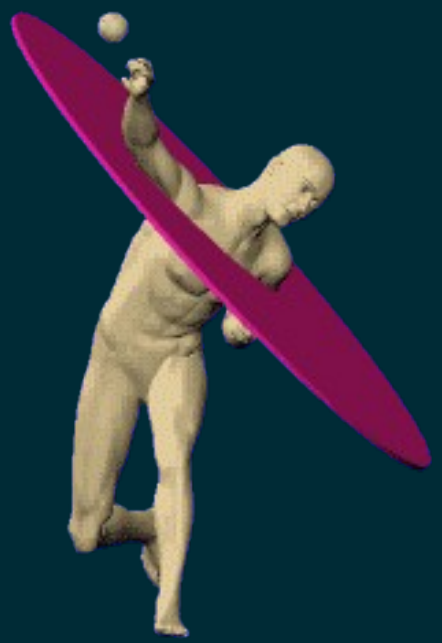


$$\text{range utilization} = \frac{\text{range in sequence}}{\text{full range}}$$

df = degrees of freedom

67 features

Biomechanics Features (Momentum)



from Nyman (2008)

Body segments:

- Rear Leg
- Lead Leg
- Glove Arm
- Torso
- Upper Arm
- Forearm
- Hand

Summary stats:

- tilt of rotation plane
- angular momentum projected onto plane of pitching hand (min/max, at release)
- angular momentum orthogonal to pitching hand plane (min/max)

67 features + **72** features = 139 features

Biomechanics Suspects

1. Max Elbow Varus Torque
2. Arm Slot
3. Min Shoulder Horizontal Abduction (aka scap loading)
4. Max Glove Arm Momentum (projected onto pitching hand)
5. Max Torso Momentum (projected onto pitching hand)

Model: multivariate proportional hazards regression

- include demographic and significant pitch-tracking features

Biomechanics Results

Feature	Effect*	95% Conf. Int.
Max Elbow Varus Torque	+2%	(–8%, +14%)
Arm Slot	+7%	(–7%, +22%)
Min Shoulder Horizontal Abd.	–4%	(–14%, +6%)
Max Glove Arm Momentum	**–21%**	**(–29%, –12%)**
Max Torso Momentum	+3%	(–8%, +16%)

*corresponding to 1-standard-deviation increase in feature

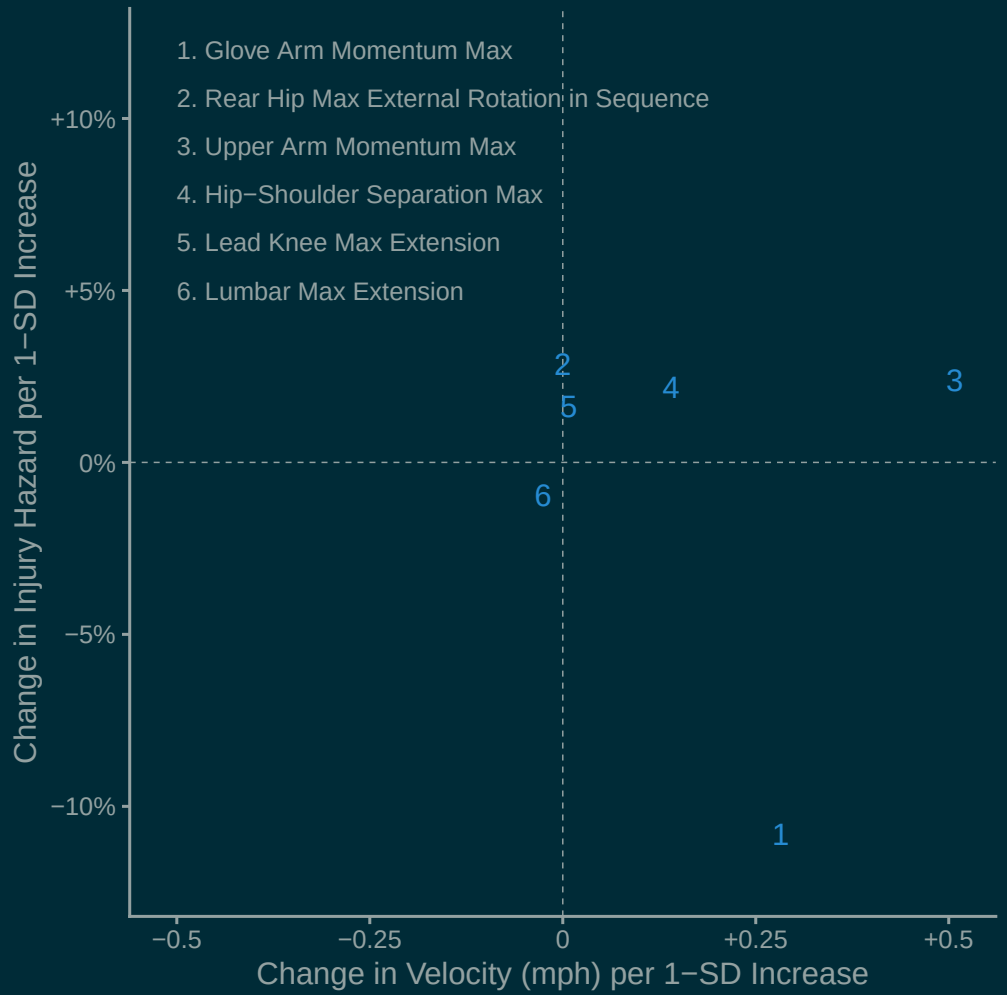
Data Mining

Model: lasso regression

- include demographic and significant pitch-tracking features
- include all 139 biomechanics features
- ℓ_1 **penalty** achieves variable selection

Two outcomes: (1) pitches until IL and (2) fastball velocity

Data Mining Results



Glove Arm Momentum Max is the only metric associated with **increased** velocity and **decreased** injury hazard

What have we learned?

First public study to estimate pitcher injury survival model using pitch-tracking data (**20-50%** injury probability over full season)

Non-fastball usage (especially **slider/sweeper**) is positively associated with injury hazard in both MLB and minors

Glove arm momentum is strongly predictive of reduced injury hazard (and increased velocity), warranting further research

Notable negative findings

- Primary Velo Effort Ratio
- Spin Rate and Glove-Side Break

Bayesian Changepoint Detection

Project led by **Rose Graves** (Rice Stats PhD student)

Goal: Identify when pitchers are pitching fatigued

Problem:

- Measurements are noisy
- Fatigue onset by game likely similar but not identical

Solution: Bayesian Hierarchical Changepoint model

1. Hierarchy across games within pitcher
2. Hierarchy across pitchers within league

Changepoint Causal Inference

Project led by **Amanda Glazer** (University of Texas)

Goal: Estimate causal effect of velocity increase

Problem:

- Rolling “enrollment” in treatment
- No treatment labels in the data

Solution:

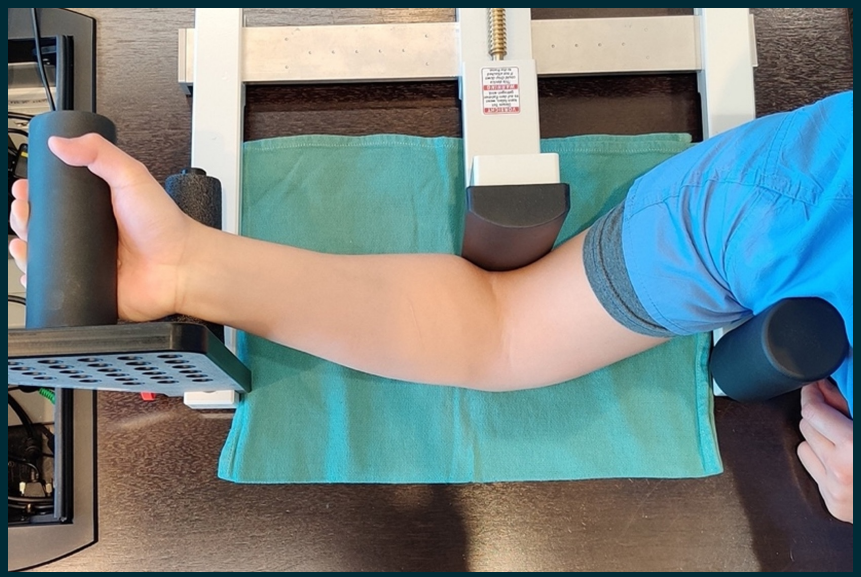
1. Identify statistically significant changepoints
2. Perform rolling treatment-control matching

Shear Wave Elastography

Collaboration with **Dr. Robert Jack** (Houston Methodist)

Problem: Not enough pitching injuries!

Solution: Measure laxity of UCL after pitching (pilot study)



from Gupta et al. (2023)

Thank You!

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